

Tintswalo Macebele

(0) 68 387 4813 | macebelekingsley@gmail.com | linkedin.com/in/tintswalo-macebele | Portfolio/tintswalo-macebele.github.io

SUMMARY

Mechatronics Engineering graduate from the University of Cape Town with experience in project engineering, warehouse operations, inventory management, warehouse floor-plan and racking design, process optimisation, and data-driven improvement. Skilled in Python, Excel, Visio, MS Project, and MS Dynamics 365 ERP, with a strong ability to analyse complex problems, improve processes, and translate operational requirements into practical engineering solutions. Demonstrated experience in developing end-to-end warehouse layouts and process flows, optimising inventory positioning, and supporting operational efficiency initiatives. Seeking challenging opportunities in operations, project engineering, and data analysis where I can apply my engineering and analytical skills to deliver innovative, high-impact solutions.

TECHNICAL SKILLS

Python, C/C++, MS Excel, MS D365, MS Visio, MS Project, MATLAB, Git, KiCad

PROJECTS

Variable filter for Grid-connected inverter system

June 2024

Electrical Engineering Project

Matlab, Python, Excel, Magnetics

- Designed a grid-connected and standalone inverter models with proposed filter design in MATLAB/Simulink
- Analysed the simulation data and successfully optimised the filter's parameters to meet the IEEE519 standards.
- Achieved state-of-the art filter design and performance for grid-connected inverter integrations

Environmental monitoring system for raptor nesting sites

May 2023

Wireless sensor network (WSN) Project

Electronics, C/C++, System Design, BOM, PCB & PCBA

- Developed the sensor node (SN) subsystem on the PCB for data retrieval and transmission to the AWS S3 bucket
- Tested and validated the firmware for reliable data transmission in Arduino IDE
- Successfully transmitted data from the SN to the database

EXPERIENCE

Project Engineer, Operations – Graduate

June 2025 – Aug 2026

Motus Aftermarket Parts

Johannesburg, SA

- Led cross-branch continuous improvement initiatives, applying Lean manufacturing to optimise warehouse operations and increase in daily throughput across 7 distribution sites.
- Analysed operational performance data and developed reports to identify process inefficiencies and drive targeted improvement initiatives.
- Designed end-to-end warehouse layouts and process flows for new sites, applying bin optimisation and strategic product grouping – improving OTIF performance by 15% and service delivery.
- Developed replenishment BSOPs to standardise inventory flow and reduce stockouts/delays.
- Improved inventory positioning using Pareto analysis principles, reducing picker travel distance by 20% and increasing accessibility of fast-moving inventory.
- Managed outsourced pallet storage capacity, coordinating product selection and quantities to optimise space utilisation while maintaining onsite efficiency for high-turnover stock.
- Implemented a cross-sell and product affinity analysis strategy for better inventory layout in the warehouse

Mechanical Engineering Vacation Student

Nov 2021 – Dec 2022

Cape Peninsula University of Technology

Bellville, SA

- Gained hands-on training in arc welding, fitting and turning (lathes), robotics, induction machines and wiring diagrams, and pneumatic systems during the CPUT 6-week programme, strengthening practical engineering skills alongside academic theory.

EDUCATION

University of Cape Town

Bachelor of Science in Engineering in Mechatronics

Cape Town, SA

March 2025

- Moshal Scholarship Award Recipient
- UCT Leadership Award for being a tutor and mentor to 2nd year engineering students
- **Relevant coursework:** Project management, Computer science

Kwa-Bhekilanga Sec School

Matric Certificate

Johannesburg, SA

Jan 2019

- Vincent Tshabalala Award for outstanding achievement in 2018 Matric Exam
- 5 Distinctions incl. Maths and Science
- **Relevant coursework:** Mathematics, Physical Science, English